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RESEARCH ARTICLE

Parametric design and three-dimensional printing: enabling Occupational therapists to develop custom hand grips

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ABSTRACT

Purpose: This study evaluates a parametric design tool created to support occupational therapists (OTs) in designing personalized assistive technologies (ATs). The tool enables users to modify existing three-dimensional (3D) handgrip models by integrating individual hand measurements and anthropometric data.

Materials and Methods: We utilized an iterative prototyping approach to develop a parametric interface for the customized design of ATs and conducted user studies involving 12 OTs to assess the practical application of the tool and gather feedback on 3D-printed ATs. We investigated the properties and parameters of 3D printing, including the temperature, layer height, infill percentage, and material selection, to enhance the safety, durability, and hygiene of 3D-printed assistive devices.

Results: The parametric design tool received positive feedback from OTs for its ease of use, customization options, and real-time design capabilities, which they found beneficial for their professional practice. The integration of this tool with 3D printing technology has also been praised for its cost-effectiveness and efficiency, offering a unique solution for customized assistive devices that address current market limitations.

Conclusion: This study introduces an innovative design tool that enhances adaptability and specialized design in AT development by integrating digital fabrication into OT practice to create personalized devices tailored to individual needs. Our study highlights the importance of precise 3D printing parameter optimization to ensure that assistive devices are both functional and reliable. Ultimately, this study aimed to enhance activity performance, such as eating, for individuals with hand impairments, thereby supporting their continued independence in daily activities.

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Assistive technology; parametric software; occupational therapy; hand impairment; 3D printing; adaptation; design tool

IMPLICATION FOR REHABILITATION

This research evaluates a parametric design tool created to support occupational therapists (OTs) in designing personalized assistive technologies (ATs). The platform enables users to modify existing 3D models of hand grips by integrating individual hand measurements and anthropometric data. The study demonstrates:

- The introduction of an efficient method for occupational therapists to devise customized grips tailored to the unique health conditions, hand measurements, and specific requirements of their clients, thereby offering a pragmatic solution that is both time-saving and cost-effective.
- The exploration of the potential of parametric design software to facilitate 3D printing technologies within clinical settings, thereby expanding the toolkit available to healthcare professionals for creating assistive devices.
- The provision of adaptable and personalized grip solutions for assistive technology users, which have the capability to significantly improve their ability to perform daily activities and foster greater independence.

Introduction

Assistive technologies (ATs) enable individuals with disabilities to participate in activities of daily living (ADL) that would otherwise be unattainable for them [1]. In individuals with hand impairments owing to aging and various pathologies, reduced hand mobility and grip strength can significantly impact ADL performance, ultimately restricting personal independence [2]. The hands play a critical role in a wide range of daily activities, including simple object manipulation and intricate motor tasks such as grasping, writing, eating, and dressing. ATs enhance the motor functions of users [3]. Therefore, to mitigate the negative effects of hand impairments, occupational therapists (OTs) frequently prescribe ATs to enhance hand mobility or prevent further damage caused by compensatory body movements [4]. ATs can be acquired either through commercial off-the-shelf solutions or by tailoring them for specific individual requirements [5]. However, purchasing off-the-shelf ATs presents a significant challenge when it comes to finding devices that meet the personal needs of users [6]. To address this issue, OTs may refer to databases such as EASTIN [7], Walmart, or Amazon to locate commercially available devices.

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Nevertheless, it is crucial to recognize that mass-produced ATs are primarily designed to meet the needs of the general population and may not be suitable for individuals with specific and unique requirements [8,9]. In addition, the abandonment rate of ATs and the reasons for abandonment are crucial factors that OTs must evaluate when considering the appropriateness of a particular device. Common reasons for abandonment include poor fit and inadequate functional requirements [10]. To address these issues, OTs often utilize various materials such as foam, tape, Velcro, cardboard, string, thermoplastic, and sponge to customize off-the-shelf or everyday objects, aiming to improve fit and enhance usability for individuals with specific needs [11]. These materials are particularly useful for quickly modifying mass-produced ATs because they are readily available and do not require OTs to acquire new skill sets [12]. For instance, foam and sponge are affordable and readily available. Their stretchability enables them to fit on various grips; however, they are not washable. Thermoplastic is a suitable material for rapid prototyping and customization, offering an optimal fit for the user's hand. However, it is not resilient to high temperatures and can be relatively more expensive compared to other materials.

To improve the durability and extend the lifespan of augmented ATs, affordable three-dimensional (3D) printing technology offers a promising avenue for the rapid fabrication of a wide range of adaptations [13,14]. These 3D printed adaptations are more durable, functional, and aesthetically pleasing compared to traditional field-modified products [15,16]. Moreover, 3D printing technology eliminates the need for tooling, making the rapid prototyping process more cost-effective [17] and time-efficient compared to traditional manufacturing methods. This is particularly beneficial for producing low-volume, highly customized parts that offer equivalent functionality [18]. Furthermore, ATs produced through 3D printing can be further customized to meet specific user requirements. This is achieved by employing a heat gun to soften the material, thus rendering it malleable, smooth, and capable of being reshaped, or by utilizing a 3D pen to append additional elements or contours to the existing printed structures [19]. However, the fabrication of an object using a 3D printer requires the development of a corresponding 3D digital model. One method involves generating these models using a 3D scanner, which captures real-world objects directly. However, the accuracy of the scanning process and the requirement for multiple scans to produce a noise-reduced model can affect the effectiveness of the adaptations. Additionally, modifying scanned objects on a computer can be challenging for individuals with limited experience [20]. Another method involves using professional computer-aided design (CAD) software to create digital 3D models. Unfortunately, many CAD tools are expensive and can be difficult for inexperienced users to navigate. Popular software such as AutoCAD or SolidWorks often present a steep learning curve, which can be a significant hurdle for OTs [21].

While some free predesigned ATs, such as Thingiverse [22,23], are available on online platforms, the selection is often limited and may not cover the diverse needs of all users. Moreover, even after downloading these models, OTs must still develop CAD skills to make necessary modifications. Despite these challenges, research indicates that OTs generally maintain a positive attitude toward adopting 3D printing technology to create ATs [24]. The goal of developing CAD software is to simplify the design process for therapists [25]. To better support OTs in AT adaptation, it is crucial for CAD tools to be closely aligned with OT clinical practice [26]. The CAD tools employed by OTs during routine operations must be highly specialized and require dedicated interactive operators. These operators must include specific features pertinent to AT adaptation such as custom expansion, custom extension, augmentation, mounting, and grasping/holding [12]. Furthermore, the software should incorporate a parametric field and labels that utilize terminology that is more consistent with OT jargon to minimize the confusion stemming from CAD-related vocabulary [27]. By integrating these specialized operators, the design and modification processes of ATs can be significantly streamlined. Ideally, the platform designed for this purpose should feature an intuitive user interface that consolidates all the aforementioned operators.

OTs' approach to AT adaptation

The AT design or modification process by OTs begins with conducting comprehensive clinical assessments to identify the needs, barriers, and functional abilities of a patient. OTs often explore existing off-the-shelf AT solutions to see if they can meet the patient's needs. If off-the-shelf ATs do not fully resolve the patient's problems, OTs assess and identify the need for adaptation by observing the usability of the ATs [2,3]. The adaptation phase follows, in which OTs start with lightweight adaptations using everyday materials they can find around them, aiming to make minimal changes to find a suitable solution. For example, modifications could be as simple as adding a piece of foam to a utensil. For more complex adaptations, OTs investigate available materials and resources to determine the appropriate approach. They were asked questions such as: "Do we need something solid? Do we need it heavy? Some people do better with weighted objects, etc." [3]. Similarly, OTs consider material properties such as whether they are bendable, moldable, or can be melted. Careful consideration is given to ensure that the adaptation does not decrease another area of use. Sharing knowledge and resources within the community is particularly important, as it allows OTs to troubleshoot and find solutions, specifically for novice OTs and those working in home care settings [28].

Materials and methods

First prototype iteration

We developed a parametric design tool using two software components: Grasshopper and Human UI. Grasshopper is a parametric modeling software that operates in conjunction with Rhinoceros, whereas Human UI is an interface paradigm specifically designed to customize the user interface of Grasshopper. In the initial iteration of the tool, we created two types of add-on grips: a round smooth grip and a round grip with a straight-line textured surface. The interface of this iteration offers a comprehensive list of customizable elements, including external dimensions, such as diameter and height, as well as internal connecting channel dimensions. For the round straight-line textured grip, we incorporated additional adjustable features such as the number of teeth, tooth length, and tooth chamfer angle, in addition to the previously mentioned parameters. We conducted a pilot test of the tool with five OTs, which provided valuable insights for further enhancement. We organized the design requirements based on the feedback from the OTs and continued to refine the interface.

Second prototype iteration

The second prototype provides various grip types, adjustable dimensions, weight options, and material selections to meet the

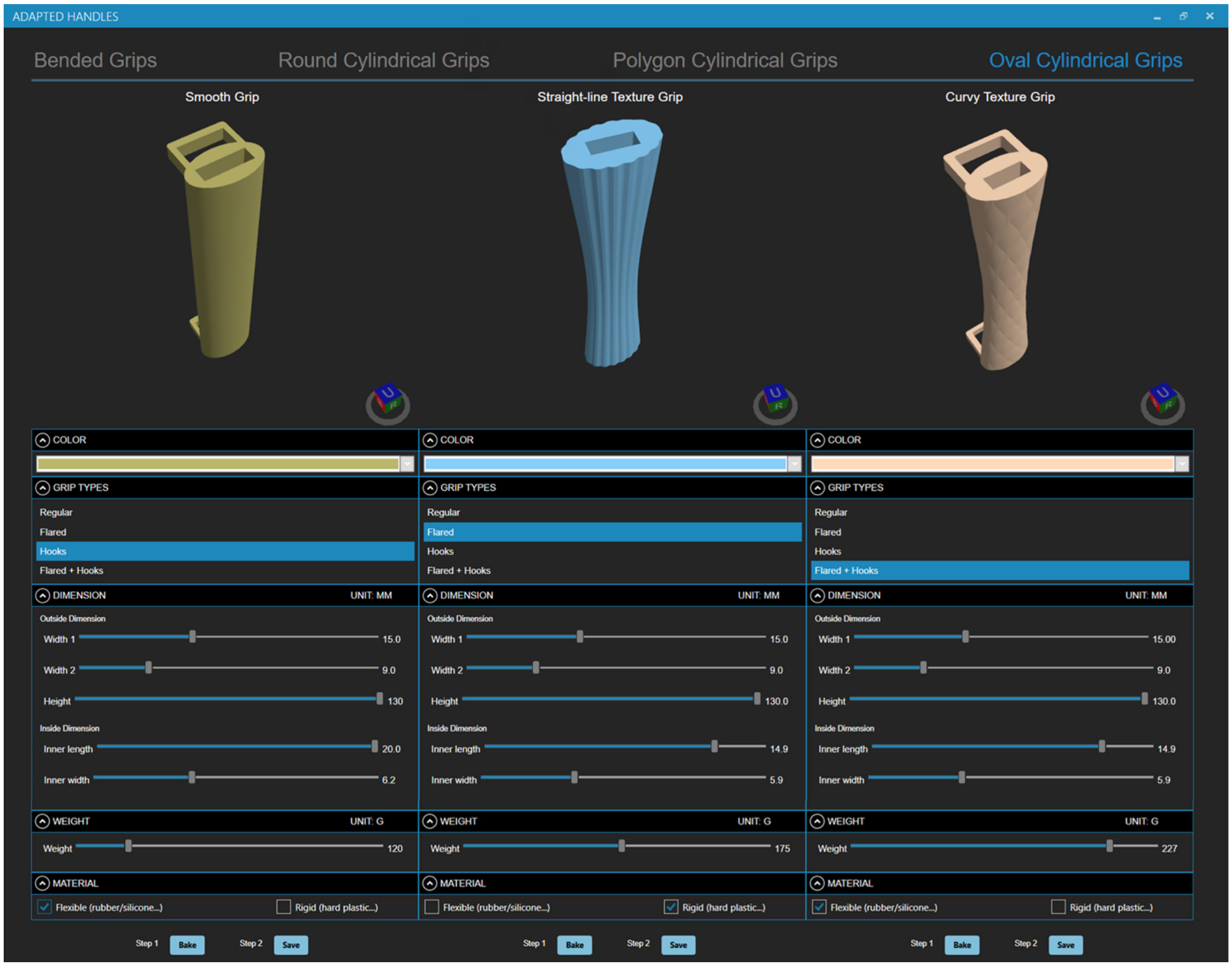

Figure 1. Parametric design tool for customizing AT grips, offering 48 designs with adjustable dimensions, weight, and materials to meet diverse hand impairment needs.

Table 1. Demographic information of OTs who participated in the user study.

Participant	Gender	Years as OT Professional	Education/certification
OT1	F	21	OTR/L
OT2	F	5	OTR/L
OT3	M	30	OTR/L
OT4	F	11	OTR/L
OT5	M	7	OTR/L
OT6	F	15	OTR/L
OT7	F	2	OTR/L
OT8	F	3	OTR/L
OT9	F	8	OTR/L
OT10	F	16	OTR/L
OT11	M	13	OTR/L
OT12	F	27	OTR/L

diverse needs of individuals with hand impairments (Figure 1). This tool includes 48 distinct adapted grip designs, each tailored to address the specific requirements of users. OTs have identified hand grips for ADLs as among the most commonly used and requested by individuals with hand impairments [12]. The general process for OTs using the parametric tool for AT adaptation is as follows. OTs first measure the user's hand to gather specific dimensions and determine the appropriate size for the utensil being adapted. They then select the desired grip shape and surface texture. They can also choose a specific color, add features (e.g., hooks), and adjust the size of the adapted grip within the parametric design tool using a slider scale based on weight while modifying the material according to each user's individual needs and functional requirements. The grip designs are informed by anthropometric Tables [29] and consider the range of motion and strength norms of the intended user. Once the design is complete, OTs can simply click "Bake" and "Save" on the interface to download the model directly to their local computer. Those skilled in 3D printing may opt to slice the model using specialized software (e.g., UltiMaker Cura) to prepare the file for printing. Alternatively, OTs with limited access or expertise may seek assistance from technicians or outsource the task to a 3D printing service provider or fabrication lab. The printer estimates the duration of the printing process and, upon completion, the OTs can collect the finished adaptive device to conduct trials with their clients to ensure its suitability and comfort.

User study

Twelve OTs, identified as OT1 through OT12 in the following sections, participated in the user study to evaluate the usability and effectiveness of the parametric design interface. All participants have experience working with individuals with hand disabilities

(Table 1). This study was approved by the Institutional Review Board (IRB) at Georgia Tech.

Procedures

The evaluation process consisted of two phases aimed at assessing both user interaction with the parametric design tool and the practicality of the resulting 3D-printed devices. In the first phase, which lasted between 37 and 75 min, averaging 43 min, the OTs used the parametric design tool to create customized grips tailored to specific user needs. They were encouraged to verbalize their thoughts while navigating the interface. Before this session, participants completed a detailed Hand Therapy Evaluation Profile that included user-specific information such as demographics, medical history, occupational details, and various physical assessments. These comprehensive profiles served as reference guides for OTs, allowing them to design adaptive devices using the parametric tool by considering the user's pathology, individual requirements, and hand dimensions based on previous patient cases (Figure 2). After the design session, semi-structured interviews were conducted to gather qualitative feedback on the usability and usefulness of the tool. Verbal responses were recorded and transcribed, and images of the adapted grip designs were collected from the interface. Once the session was completed, the AT devices conceptualized by the OTs were fabricated. The printed adaptive grips were then provided to the OTs for evaluation in a follow-up session. The second session reconvened the original participants and lasted between 15 and 20 min, with an average duration of 17 min. During this time, the OTs engaged in interviews regarding the design and practicality of adaptive AT devices. Upon completion of both phases, the qualitative data collected from the OTs were transcribed and analyzed using thematic analysis to extract insights into the interface and 3D-printed ATs.

Investigating 3D printing parameters

Optimizing 3D printing parameters is crucial for creating high-quality, functional ATs, with safety, durability, and hygiene as the primary considerations for OTs. To address these issues, understanding the performance and ideal printing techniques for different materials (filaments) is vital. We explored three widely used 3D printing materials—polylactic acid (PLA), polyethylene terephthalate glycol (PETG), and thermoplastic polyurethane (TPU)—utilizing cylinders that represent adapted grips to visualize the key parameters and demonstrate the outcomes of each material.

Commercially available 3D printing filaments often specify a wide temperature range, from 15°to 40°. The minimum, middle, and maximum temperatures for each material were tested to determine the optimal conditions. Additionally, the costs of the materials differ: 1 g of PLA costs approximately $0.022, 1 g of PETG also costs $0.022, and 1 g of TPU costs approximately $0.124, with TPU generally being more expensive than the other two materials. The test samples, which were one-inch diameter cylinders with a height of one inch, were printed using UltiMaker 3/S3/S5 printers. All settings were adjusted using the Cura slicing software, and we recorded the time, cost, and status of each print.

Results

Usability and functionality of the tool

After exploring the software, the OTs recognized the usability of the parametric design tool from multiple perspectives and provided positive feedback on its usefulness in their professional practice. Highlighting its ease of use, OT4 stated: "*It seems like it would be easy to use if you had the dimensions just to input*" [OT4]. OT7 remarked, "*I found it pretty self-explanatory. If it is for someone who would use this, I think they would probably intuitively understand its functions. It was pretty straightforward to use*" [OT7]. OT1 added, "*I liked that I was able to turn it like in all 3D shapes… pretty quick that you can learn… I just need to slide the slider*" [OT1]. Similarly, OT5 noted, "*It's pretty easy to use… from the types (adapted grip) to the dimension to the weight to the material, that's pretty clear*" [OT10].

Six OTs acknowledged various aspects of the tool's functionality. OT5 and OT10 appreciated the variety and customization options, with OT10 stating, "*I see there's so many grips! I would be so excited to get to choose from these for her*" [OT10]. The ability to alter colors and view changes in real-time was highlighted by OT1, OT3, and OT8. OT3 said, "*Having the option of changing colors is great, because some people may have low vision…*" [OT3]. OT8 emphasized, "*For dementia, you want to use colors like red… it grabs their attention. They need cues to eat, like reminders to eat their food*" [OT8]. The significance of real-time monitoring in 3D design was also emphasized by OT4: "*Manipulate it like from a position standpoint. So that's actually cool and that would help like if I was fabricating something for sure*" [OT4], and OT10 added, "*Drag and*

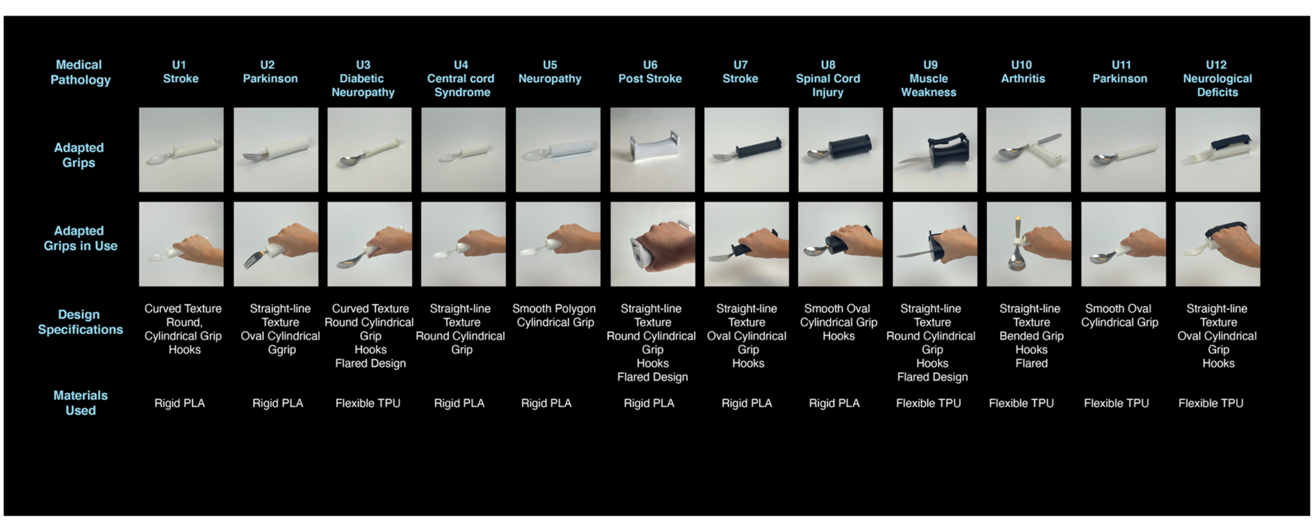

Figure 2. 3D-Printed ATs were designed by OTs, considering users' specific needs identified through a detailed hand therapy evaluation profile.

zooming function is very cool. What a neat interface!" [OT10]. Additionally, the participants highlighted the ease of inputting precise measurements. OT4 noted, "*Double click the bar to type the number; that's easier*" [OT4]. OT10 echoed this, stating, "*I like that you can double click the slider to type in the number*" [OT10]. Furthermore, OT1 emphasized the tool's utility in aiding decision-making, saying, "*You have a little line that says like this is more for Parkinson's maybe the cylindrical grasp that's helpful*" [OT1].

Usefulness and uniqueness of the tool

The OTs stated that the tool would be beneficial for their everyday practices. OT7 stated, "*That's cool, I think in real life I would definitely use this*". OT10 elaborated, "*This tool exceeded my expectations… I love this idea and it is a need… How can we help you be more independent? That's our question for our client. And then we could say, here's another tool in our toolkit to help you do something better…And it was designed by your therapist and you. I think that is very valuable*" [OT10]. OT10 also envisioned how this tool could be integrated into their practice: "*I see that being a very positive way to build toward the future. The company could have it be part of our existing system, just a built-in click on that link, because we're always connected to the internet wherever we are*".

The OTs emphasized the uniqueness of the tool in addressing current challenges in assistive device provision. OT5 stated, "*Because it's a very customized design, it has a big benefit for all the patients. We have so many standard-sized utensils on the market, making it hard and expensive to give a patient a try*". OT8 added, "*What you're doing is cool because we basically gave a one-size-fits-all… like this foam tubing, and it really didn't fit all, but it's all we had*" [OT8]. OT6 remarked, "*I think customization really differentiates your products from others, and that makes it unique on the market*" [OT6]. OT9 noted, "*You give me an opportunity to create a specialized adapted grip just made for him… as opposed to buying a universal one from Amazon, which can be hard to adjust at times*" [OT9]. OT8 also highlighted the potential of the tool for various user groups, specifically in pediatric settings: "*For instance, this would be a good tool… they (pediatrics) can't use typical silverware… this will be a cool tool because those types of utensils are typically nothing like the radius of your typical utensil*" [OT8].

Cost-Effectiveness and efficiency of 3D printing

Six OTs expressed positive feedback regarding the integration of 3D printing technology into their practice. OT2, OT3, OT4, and OT11 emphasized the cost-effectiveness of the tool, noting the high price of conventional products. OT2 stated, "*It costs $15 to $20 for just one spoon, which can be really expensive for many patients. I believe 3D printing could reduce the cost of producing such items*". OT4 noted the affordability of producing multiple versions, stating, "*It's cheap. So you could get three or four different versions*". The time efficiency of 3D printing was considered acceptable, with an average adapted gripping time of approximately six hours. OT5 commented, "*I mean, they wait for weeks and weeks on a wheelchair cushion, so as long as they had it by their next session…*".

Reflections on the 3D printed adapted grips

All the OTs expressed positive feedback on the practicality and utility of the adapted grips they designed. OT5 remarked, "*Because she had arthritis on the hand… she cannot make a full fist, so sometimes she could not hold a regular utensil. That's why this one perfectly matches her hand. And it has a hook, it makes gripping more secure*". OT8 appreciated the distinctiveness of the grips, commenting, *"It is impressive because I have never encountered a grip of that size, which is what users need"*. OT10 noted, *"The adapted grip is very useful because they cannot twist their wrist, so they have to compensate. The most important thing is to improve their self-independence in completing the task of feeding"*. Six OTs confirmed the effectiveness of the adapted grip patterns. OT4 commented, "*The curvy texture could help with traction*", and OT9 noted, "*The straight-line texture here is good; it is not sliding easily*". Three OTs emphasized the grips' versatility for various utensils. OT3 noted, "*I have enabled her to have two kinds of grips… It's more of a universal grip*". OT11 added, "*I could see it translating quickly to other self-care utensils*". OT3 emphasized the hygienic benefits, stating, "*The adapted grip is very easy to clean… hygiene is very important for eating utensils*", aligning with OT8's perspective: *"Cleanliness is the most important thing"*.

Enhancements for adapted grip design

Several OTs provided valuable suggestions for enhancing the 3D models of adapted grip designs. OT7 recommended expanding material options beyond the rigid and flexible categories, stating, "*Having more material options might be helpful instead of just rigid and flexible*". This corresponds to the options provided by different 3D printing technologies, including soft PLA, stereolithography, and metal printing. OT9 suggested adding a dot texture to provide extra sensory input for users with diminished sensation, proposing "more little dots on the surface". Additionally, OT7 proposed altering the inner design for more functional versatility, questioning the rectangular shape: "*I wonder how functional that is, being a rectangle instead of like an oval*". OT8 suggested enlarging the slot for hooks, stating, "*I am trying to think about maybe making that slot (hooks) a little bigger*".

Results of 3D printing parameter optimization

The results of our exploration of 3D printing methods suggest that the performance of each material varies with temperature, which affects both the surface quality and internal structure. PLA is a versatile and affordable material known for its durability and safety; for this project, we used Matter Hackers white 2.85 mm PLA. The optimal printing nozzle temperature for PLA ranges from 180 °C to 220 °C. Results indicate that printing at 200 °C yields the smoothest surface and most uniform layers, while the sample printed at 180 °C shows noticeable gaps. In contrast, PETG is a rigid, food-safe material recognized for its exceptional durability and thermal stability, with recommended printing temperatures between 220 °C and 260 °C. The results indicate that the sample printed at 240 °C has the shiniest surface; however, the print at 220 °C provides a smoother finish and more uniform top structure. Lastly, TPU is a flexible material with a recommended printing temperature range of 225 °C–240 °C. The sample printed at 225 °C exhibits the cleanest surface; however, higher temperatures result in stickiness and increased internal wadding (Figure 3). We also experimented with different layer heights, infill percentages, and wall line counts, showing their effects on print quality, time, and material usage (Figure 4).

Discussion and contribution

Enabling digital fabrication in OT practice

The traditional process of prescribing ATs often involves a time-consuming trial and error approach, sometimes requiring

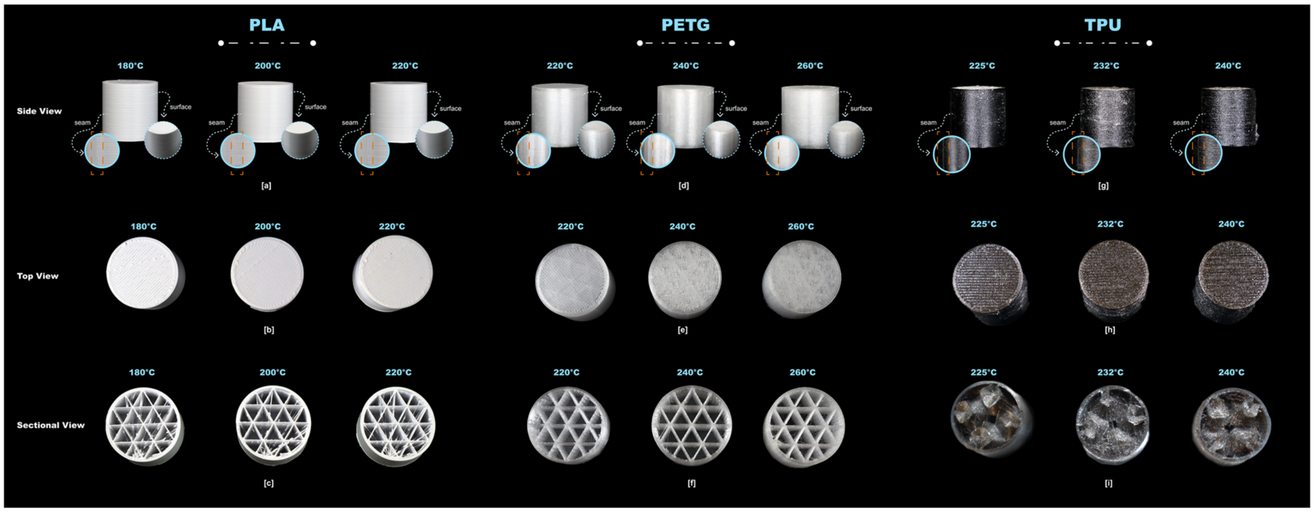

Figure 3. Comparison of the 3D printing parameters for PLA, PETG, and TPU, highlighting their optimal printing temperatures, resulting print qualities, and differences in internal structures.

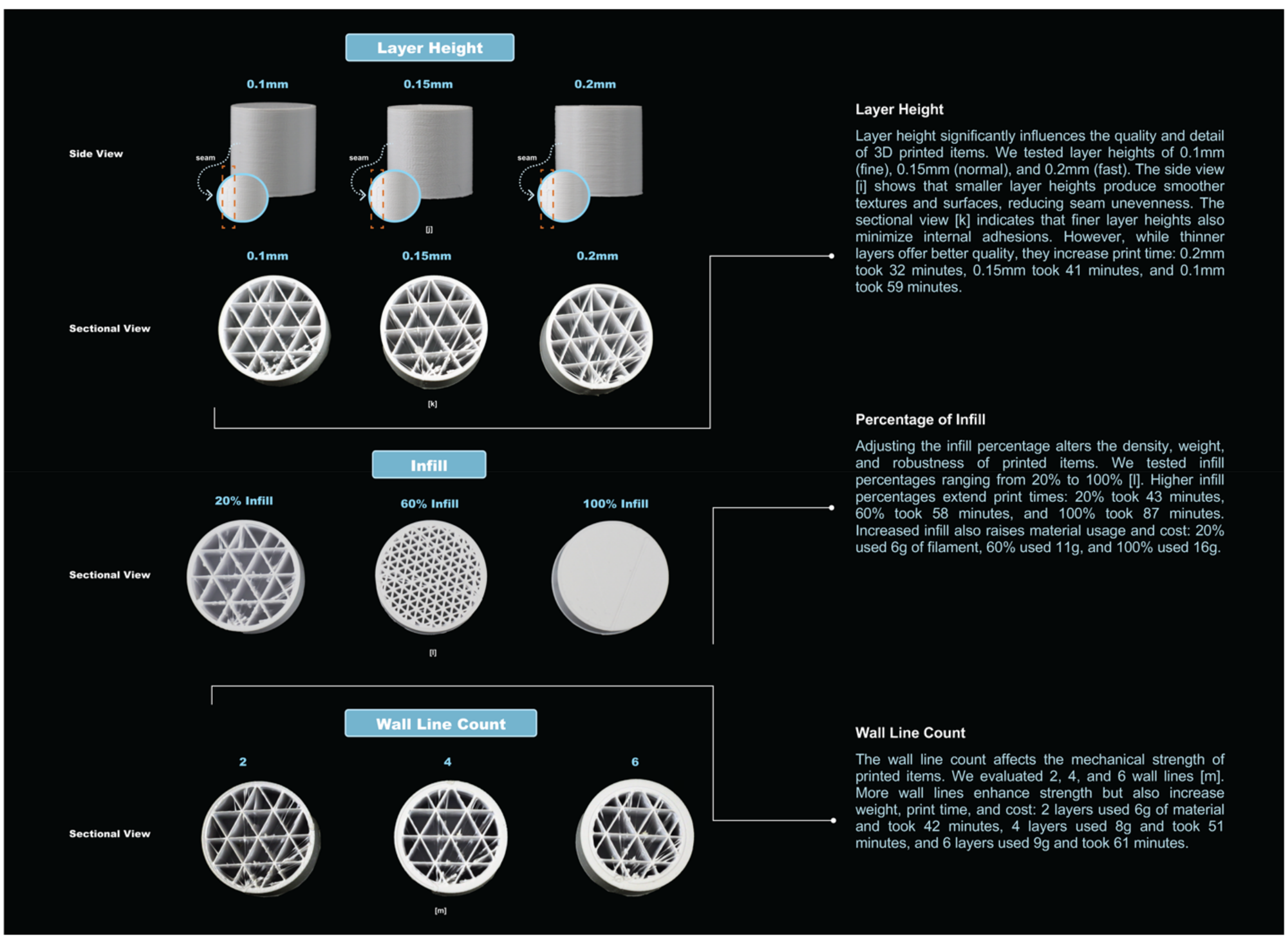

Figure 4. Comparison of 3D printing parameters for PLA, PETG, and TPU, focusing on layer height, infill percentage, and wall line count, showcasing their effects on print quality, time, and material usage.

days or weeks of waiting for devices [30]. However, integrating 3D printing with parametric modeling software can significantly accelerate this cycle. If an adapted grip requires redesign, a new, customized version can be produced quickly, streamlining the entire process and reducing wait times. The improved durability of 3D printed adaptive devices also offers a more sustainable solution, optimizing replacement costs and minimizing the time required to prescribe new ATs [31].

OTs can refine modified ATs using tools such as heat guns or 3D pens, leveraging their existing skills to make adjustments. This flexibility allows for further customization, enabling patients and OTs to personalize assistive devices with unique patterns, enhancing aesthetic appeal—a crucial factor in AT acceptance [32]. Moreover, the production cost of a single adapted grip using 3D printing (ranging from 1 to 5 dollars) is significantly lower than that of commercially available ATs. The decreasing price of 3D printers further supports this affordability, making it feasible for OTs to adopt 3D printing technology for AT device creation, as noted in previous research [4,24].

Enhancing design adaptability in AT design

Our parametric software provides flexibility in crucial variables, leveraging OT expertise to tailor adapted grip designs for specific pathologies, hand measurements, and individual needs, thereby resulting in personalized solutions that benefit both AT users and therapists. Previous studies have investigated various methods for improving the digital production of customized adaptive devices, including the use of AutoCAD and integration of 3D scanning [33]. While earlier tools faced limitations in terms of usability and alignment with OT requirements [27], our proposed parametric design tool offers an innovative approach to 3D modeling that effectively incorporates digital fabrication into the routine creation of adapted grips by OTs. It emphasizes user-friendliness, time efficiency, and compatibility with OT practices, requiring minimal training for software use. The streamlined design process simplifies the fabrication workflow, provided a 3D printer is available.

Supporting specialized design of ATs

Previous research has highlighted the limitations of universal design approaches in developing adaptive devices such as adapted grips for conditions such as Parkinson's disease [34]. Although some devices offer weight customization, fixed dimensions continue to restrict their functionality and inclusivity. The specialized design approach to AT development is essential for addressing the unique needs of individual users, diverging from universal design paradigms [35]. Interviews with OTs revealed a tendency to prescribe standard off-the-shelf hand ATs owing to various constraints. Commercially available ATs are usually offered in a limited range of sizes, reflecting universal design principles that aim to create widely accessible products and environments [36]. However, this approach often results in high abandonment rates owing to poor fit for individual users' unique body specifications [37]. In contrast, the parametric modeling tool could significantly advance the field by facilitating the development of adapted devices that meet specialized design objectives and cater to the unique abilities and requirements of individuals.

Key considerations in 3D printing ATs and material properties

Optimizing 3D printing parameters is essential for producing high-quality, functional ATs, with safety, durability, and hygiene being key concerns for OTs. Although mechanical and morphological tests are typically not applied to smaller 3D-printed items that do not encounter regulated forces, designers and rehabilitation engineers must assess various factors related to manufacturing, 3D printing, and material properties. The choice of materials and filament is crucial because different options possess unique characteristics. For instance, while joysticks for wheelchair users typically experience minimal force, heat-resistant materials are crucial for components utilized in vehicles. Adjustments to layer height, orientation, and thickness can significantly affect the final product. Although minor changes in these parameters may result in minimal differences, increasing the wall count and infill generally improves strength. Temperature settings also affect the finish of the printed item, with cooler temperatures yielding a matte finish and higher temperatures producing a shinier appearance. Additionally, the orientation of the print significantly influences the strength of the final product because there is always a risk of parts breaking during printing. Notably, printing time and cost can vary based on print quality; therefore, factors such as higher infill or thicker walls do not always guarantee an acceptable outcome for OTs in rehabilitation settings because of the limited time and resources dedicated to such work.

Conclusion and future work

This study presents an innovative parametric design tool developed for OT practice to customize and produce ATs for individuals with hand impairments. The significance of this tool lies in its capacity to offer personalized adaptive devices, thereby enhancing the alignment between devices and user' abilities. This simplifies the process for AT providers to create tailored devices based on users' preferences, health conditions, and hand measurements, reducing time and effort compared with conventional trial-and-error methods. We anticipate that this tool will have far-reaching implications beyond individual customization, potentially transforming assistive device provision. Its adoption promises rapid, precise, and personalized solutions, ultimately enhancing the quality of life for individuals with hand impairments.

Future plans include refining the interface based on project outcomes and incorporating additional commonly used AT categories. The tool is currently being utilized in a rehabilitation center and is undergoing further development to improve its usability. Additional studies involving more individuals with disabilities, their family members, and caregivers are necessary to comprehensively evaluate the interface and physical adaptive devices. We intend to transition the tool to an online platform to make it accessible to the broader AT community.

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